Quiz ~~1~~ 2 (Version ~~1~~ 2)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not recieve credit it you don't show your work.

1 Matrix Equations

Determine a general form solution for the following matrix equation.

$$\begin{bmatrix} -1 & -4 & 0 \\ 2 & 8 & -1 \\ 3 & 12 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -2 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -3 \\ 6 \\ 1 \end{bmatrix} - \begin{bmatrix} -2 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -4 & 0 & -1 \\ 2^{-2} & 8^{-8} & -1 & 3^{-2} \\ 3^{-3} & 12^{-12} & -1 & 4^{-3} \end{bmatrix} \sim \begin{bmatrix} -1 & -4 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1^{+1} & 1^{-1} \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$x_1 = 1 - 4x_2$

x_2 is free

$x_3 = -1$

2 Full Span

Determine if the following vectors span all of $\mathbb{R}^3$. Show your work and justify your answer.

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -3 \\ -6 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & -2 \\ 1^{-1} & -3^{+2} & 2^{-2} \\ 3^{-3} & -6^{+6} & 5^{-6} \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & -2 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Solution.

Yes. 1 pivot per row

3 Linear Equations and Span

Determine if span$\{\mathbf{v}_1, \mathbf{v}_2\}$ is the same as the solution set of the given linear equation. Show your work and justify your answer.

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \qquad \mathbf{v}_2 = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} \qquad - x_1 + 2x_2 + 4x_3 = -3$$

S

$-3 + 2(-2) + 4 = -3 - 4 + 4 = -3$

$-5 + 2(1) + 0 = -5 + 2 = -3$

$\vec{v}_1 \in \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : -x_1 + 2x_2 + 4x_3 = -3 \right\}$

$\vec{v}_2 \in \{ \ \dots \ \}$

but $\vec{0} \in S$ and $\vec{0} \in \text{span}\{\vec{v}_1, \vec{v}_2\}$

Solution.

No. $\vec{0} \in \text{span}\{\vec{v}_1, \vec{v}_2\}$ but

$\vec{0} \notin$ sol. set. of $-x_1 + 2x_2 + 4x_3 = 3$